

IXORA Ecosystem — Product & Architecture Review

Data: 7 de junho de 2026 | **Escopo:** front_vibes, back_vibes, ixora-admin, ixora-infra | **Método:** Análise read-only (sem alterações de código)

1. Executive Summary

IXORA é uma **plataforma de áudio ambiente funcionalmente madura no Android**, com um loop central pronto para produção: Firebase auth → sync Laravel → CRUD de vibes → plano de execução → playback nativo → download offline. O stack de catálogo/admin (sounds, covers, preset vibes) e a infraestrutura de staging estão substancialmente implementados.

Desde o baseline de 2 de junho de 2026, as maiores melhorias foram:

- **Backend confidence sprint** — testes de Vibe e VibeSound API (~150 casos Pest, antes 104)
- **CI quality gates** — `quality.yml` nos 3 repositórios de app
- **Catálogo de sounds escalável** — search, filters, sort, pagination no backend + admin

A plataforma **não está pronta para produção** nem para **automação Scheduler end-to-end**. O Scheduler existe apenas como **stubs de banco + docs de planejamento**.

Scheduler MVP: GO condicional — iniciar spec/ADR + Schedule CRUD no backend; **NO-GO** para playback automatizado completo sem FCM e UI mobile.

2. Strengths

Área	Evidência
Mobile playback	NativeAudio + execution plan + FGS + media session + audio focus + headset disconnect
Offline audio	Dual manifest, rollback atômico, file:// playback, health assessment, QA nativo
Backend domain	Vibes, vibe sounds, sounds, covers, presets, storage — Form Requests, Resources, Policies
Auth	Firebase JWT verify, sync fail-closed, admin approval — ADR-001
Storage	Laravel-only writes DO Spaces, safe deletion — ADR-002
Documentação	8 ADRs, 12 feature specs, architecture map, scheduling-model.md
Testes	~150 testes backend; gaps de vibe/vibe-sound fechados
Staging IaC	OpenTofu: VPC, Postgres, Spaces, App Platform (API + worker + admin)

3. Weaknesses

Área	Gap
Scheduler	Schema + models + policy — zero API, jobs, UI mobile, testes
Produção	Sem OpenTofu prod; state local; sem observabilidade
Push	Só media notification — sem FCM para wake agendado
iOS	Sem projeto Capacitor iOS
Ownership UX	Enforcement só no servidor; mobile sem handling 403
Fades	Armazenados no DB mas não aplicados (ADR-008)
Admin tests	1 arquivo, 5 casos; sem E2E
QA em CI	Playwright + WDIO locais; não no CI
Rate limiting	Sem throttle na API
Deploy legado	SSH workflow coexiste com App Platform

4. Risks Before Scheduler

Risco	Severidade	Detalhe
Sem push wake	Alta	FCM/APNs necessário; sem isso só foreground
Schema imaturo	Média	Sem timezone, next_run_at, dedupe de ocorrências
Offline + automação	Média	Snapshot stale causa falhas silenciosas
Limites Android BG	Média	Task removal mata processo; prepare handshake falha
Dual deploy	Média	SSH + App Platform pode causar drift
Sem observabilidade	Média	Execuções falhas invisíveis
Recurrence/TZ	Alta	DST, idempotência, política de timezone indefinida

5. Recommended Next Milestone

Milestone: Scheduler Foundation Sprint

- 1. Scheduler feature spec + ADRs (timezone, recurrence, push, idempotency)
- 2. Backend Schedule CRUD API (controller, requests, resources, factories, Pest)
- 3. Schema hardening (timezone, next_run_at)
- 4. FCM spike (token storage, payload mínimo → playVibe)
- 5. Mobile schedule list/create UI (CRUD only, sem runtime)
- 6. Retirar deploy SSH legado

6. Updated Completion Percentages

Categoria	2 Jun	7 Jun	Notas
SSD / Specs / ADRs / Architecture	~75%	78%	8 ADRs, 12 specs; sem scheduler spec
Quality Harness / Sensors	~50%	62%	CI em 3 repos; E2E fora do CI
Foundation / Core Platform	~80%	83%	Auth, sync, health sólidos
Backend Laravel	~70%	80%	Testes vibe; API scheduler ausente
Mobile Ionic/Vue/Capacitor	~78%	81%	Android maduro; sem iOS
Infra Staging	~65%	68%	OpenTofu staging; deploy legado
Audio Engine / Player	~85%	88%	FGS, focus; fades ausentes
Offline Audio	~80%	83%	Nativo only; recovery forte
Core Vibes CRUD	~75%	88%	Mobile + API + testes
Sounds/Admin Platform	~72%	84%	Search/filters/pagination OK
Production Readiness	~35%	38%	Staging only; sem monitoring/DR
Automation System	~5%	8%	Schema + planning doc
Smart Home	~5%	5%	Schema comments only
Native Platform Architecture	~72%	76%	Android customizado; patch-package

7. Go / No-Go — Scheduler MVP

Fase	Veredito	Rationale
Spec + ADR + Backend CRUD	GO agora	Schema/models/policy existem; player + execution plan estáveis
Scheduler MVP completo (autoplay auto)	NO-GO	Sem FCM, worker, UI mobile, testes

Scheduler Prerequisites

Pré-requisito	%	Status
schedules + schedule_executions schema	~85%	Migrations OK; falta timezone, next_run_at
Models + SchedulePolicy	~70%	Existem; sem testes/factories
Vibes + execution plan (client)	~95%	Shipped
Backend Schedule API	0%	Ausente
Worker / cron / queue jobs	0%	Ausente
Push (FCM)	~5%	Não implementado
Mobile schedule UI	0%	Ausente
Scheduler tests	0%	Ausente
Scheduler spec/ADR	~15%	Só planning doc

Conclusão geral dos pré-requisitos Scheduler: ~18%

8. Mobile App — front_vibes

Authentication

Feature	Status
Firebase email login/signup	Complete
Google login (web + native)	Complete
Password reset	Partial — só envio de email
Route guards	Complete
Auth sync	Complete — fail-closed
Apple Sign-In	Missing

Vibes

Feature	Status
List / Details / CRUD / Sounds	Complete
Ownership (frontend)	Missing — só backend

Player

Feature	Status
Playback / pause / resume / stop	Complete
Media notification + lock screen	Complete (native)
Audio focus + phone interruption	Complete
Headset disconnect	Complete
Play modes (loop/once/interval)	Complete
Fade in/out	Missing (intencional)

Offline

Feature	Status
Download / storage / manifests	Complete (native)
file:// playback	Complete
Remove download / recovery	Complete
Router offline guard	Partial

QA

Tool	Status
Vitest	5 arquivos; sem player/auth CRUD
Playwright	1 spec; fora do CI
WDIO + Appium	QA nativo rico; manual/local
CI	lint + build + vitest

9. Backend — back_vibes

Authentication — Complete

FirebaseAuthenticate middleware, SyncFirebaseUser, POST /api/auth/sync. 4 sync tests + 9 unit tests. Gap: middleware não auto-sync; token válido + user ausente → 401.

Vibes — Complete

CRUD com VibePolicy, Form Requests, VibeResource. 14 feature tests. Imagens via admin/uploads.

Vibe Sounds — Complete

Attach/detach/update, sort_order, fades, timing, play modes. 15 feature tests. Gap: sem bulk reorder.

Sounds Catalog — Mostly Complete

CRUD, SoundCatalogQuery (search, category, tag, status, sort, pagination). 36 testes. Gap: lista default inclui inactive; SoundPolicy não wired em writes.

Storage — Complete

DO Spaces, upload validator, safe deletion, POST /api/admin/uploads. ~30 testes storage.

Scheduler Readiness — Schema Only (~15%)

Artifact	Existe	Wired
schedules migration	Sim	Sem API
schedule_executions migration	Sim	Sem API
Schedule / ScheduleExecution models	Sim	Sem controller
SchedulePolicy	Sim	Sem routes
Controller / jobs / tests	Não	—

Test Count

Métrica	Valor
Arquivos de teste	21
Casos Pest	~150
Bem coberto	vibes, vibe sounds, sounds, storage, presets, covers, auth
Ausente	schedules, devices, user settings, /auth/firebase

10. Admin Panel — ixora-admin

Área	Status
Authentication	Complete
Sound management	Complete
Catalog (covers/presets)	Complete
Upload flow	Complete — Laravel XHR
Search/filters/pagination	Sounds OK; covers/presets básico
Validation	Server authoritative; sem file-size client
Error handling	Complete
Tests	1 arquivo, 5 casos

11. Infrastructure — ixora-infra

Área	Status
DigitalOcean staging	Complete
App Platform (API + worker + admin)	Complete
Managed Postgres PG16	Complete (single node)
Spaces	Bucket IaC; CDN/CORS manual
OpenTofu	Staging only; state local
GitHub Actions (infra)	Missing
Production IaC	Missing

12. Production Readiness

Concern	Classificação
Security	PARTIAL — VPC, CORS, secrets; sem rate limit/WAF
Backups	NOT READY
Observability	NOT READY
Logging	PARTIAL — stderr only
Error reporting	NOT READY
Rate limiting	NOT READY
Monitoring	NOT READY
Disaster recovery	NOT READY

Overall: PARTIAL (~38%) — adequado para staging, não para produção.

13. Mudanças desde 2 de Junho

Item	Antes	Agora
Vibe API tests	0	14
VibeSound API tests	0	15
Sounds search/filters/pagination	Missing	Complete
CI quality gates	Partial	Green em 3 repos
Backend tests	104	~150
Admin tests	0	5

14. Bottom Line

IXORA tem uma **base sólida de playback ambiente** adequada para iniciar **CRUD de Scheduler no backend e trabalho de especificação**. O Scheduler MVP completo com wake-and-play automatizado requer **FCM, worker runtime, UI mobile e ADRs** — hoje ~18% dos pré-requisitos estão completos.

Gerado em 7 de junho de 2026 — Análise read-only do ecossistema IXORA. Referência: [ixora-infra/docs/ecosystem-current-state-report.md](#) (2026-06-02).